$$E + S \underset{k_2}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_3} E + P$$

If, only the initial period of the reaction is considered, the product concentration [P] is negligible and the formation of ES from product can be ignored.

The velocity of the overall enzymic process

$$v = k_3[ES] \quad ...(1)$$

Rate of formation of ES = $k_1[E]\,[S]$

Rate of breakdown of ES = $[k_2+k_3]\,[ES]$

Under steady state conditions,

Rate of formation of ES = Rate of breakdown of ES

(i.e) $k_1[E][S] = [k_2 + k_3]\,[ES]$

$$[ES] = \frac{(k_1[E][S])}{(k_2 + k_3)}$$

$$[ES] = \frac{([E][S])}{\frac{(k_2 + k_3)}{k_1}} = \quad ...(2)$$

Where, $K_m = \frac{(k_2 + k_3)}{k_1}$, is a constant known as Michaelis constant.

The concentration of uncombined enzyme [E], is equal to the total enzyme concentration [ET] minus concentration of ES complex, (i.e) [E] = [ET] – [ES]

Substitute this value of [E] in equation (2), we get

$$[ES] = \frac{(\{[E_T]-[ES]\}\,[S])}{K_m} \quad ...(3)$$

$$[ES]\,K_m = [E_T]\,[S] - [ES]\,[S]$$

$$[ES]K_m + [ES]\,[S] = [E_T]\,[S]$$

$$[ES]\,(K_m + [S]) = [E_T]\,[S]$$

$$\therefore [ES] = \frac{([E_T][S])}{(K_m + [S])}$$

Substitute this value of [ES] in equation (1)

$$v = \frac{(k_3[E_T][S])}{(K_m + [S])} \quad ...(4)$$

The velocity becomes maximum velocity (V_{max}) when the catalytic sites on the enzyme are saturated with substrate.

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